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OUTDOOR EQUIPMENT FOUNDATIONS

PROPRIETARY INFORMATION

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

This Transmission Engineering Standard (TES) intended to be used in the system of the National Grid Saudi Arabia, defines the requirements governing the foundation design of the substation buildings, transformer and other structures and equipment, and provide basis and guidelines for the Design Engineer.

2.0 CROSS REFERENCES

This Transmission Engineering Standard will always be read in conjunction with National Grid Saudi Arabia Specification No. TES-P-119.01, Latest Revision, Titled "Introduction to Substation Design Standard", which shall be considered as integral part of this TES.

3.0 APPLICABLE CODES AND STANDARDS

The latest revisions/amendments of the following Standard and Specifications are intended as guidance in providing an acceptable level of quality and practice. In case of conflict between these Standards and the text of this Standard, the latter shall govern:

3.1 Company Standards

1.	TCS-Q-113.02	Earthworks
2.	TES-P-119.09	Bus bars and Supporting Structures
3.	TES-P-119.19	Building and Yard
4.	TES-P-122.05	Transmission Structures
5.	70-TMSS-03	Normal Weight Ready-Mixed Portland Cement Concrete

3.2 SASO -Saudi Arabian Standards Organization

1.	SSA 2	Steel Bars for the Reinforcement of Concrete
2.	SSA 142	Cement Tests
3.	SSA 143	Portland Cement -Ordinary & Rigid Hardening
4.	SSA 378	Concrete Aggregates
5.	SSA 690	Test Method of Steel Bar for Reinforcement of Concrete
6.	SABC	Saudi Arabian Building Code

3.3 International Standards

1.	ACI 318M	Building Code Requirements for Structural Concrete
2.	ASTM A615	Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement
3.	ASTM D698	Moisture-Density Relation of Soils
4.	ASTM D1194	Test for Bearing Capacity of Soil for Static Load on Spread Footings
5.	ASTM D2049	Test for Relative Density of Cohesionless Soils
6.	ASTM D2487	Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System).
7.	UBC	Uniform Building Code

4.0 GENERAL CONSIDERATIONS

4.1 Design Basis for Foundation

4.1.1 All foundation designs shall conform to the requirements of International Code of Practice for Foundations and shall take into the soil conditions on site. Design calculations shall clearly show the calculation and design method of all proposed foundations, concrete structures and all ground conditions. Included in the design are applied loads, actual and allowable stresses, overload capacity factors, soil parameters and other pertinent information.

4.1.2 Foundations shall be capable of resisting all the design loadings and shall be designed to keep differential and total settlements or other movements including the transmission of dynamic loading shall be considered within the acceptable limits. The Design Engineer shall ensure that the stability of buildings and structures as a whole or any part of them and calculated factors of safety conform to the International Code of Practice.

4.1.3 The design of foundations shall take in to consideration the requirements of pipe, cable ducts, duct bank & trenches, foundation of bolts for equipment or structural components and concrete protection.

4.1.4 All foundation in reclaimed areas shall be founded on piles or ground subject to ground improvement, as recommended by the soil investigation agency.

4.1.5 Ground improvement or Piling method techniques to be used shall entirely meet the recommendations of the soil investigation agency and the requirements of design with particular reference to the peculiarity of each site and taking into the consideration the following.

a. The water-bearing nature and permeability of the strata
b. The presence of unstable sands and silts
c. The nature of the bedrock
d. The presence of contaminating chemicals in groundwater & soil
e. All other geological features

4.1.6 Where piling is adopted, the design of the piles and the foundations they support shall be to the approval of the Engineer. Where foundations are supported on piles they shall be adequately and properly tied to the pile heads.

4.1.7 Testing shall be required at each site, where piling methods or ground improvement techniques have been applied. The extent and form of such testing shall be appropriate to the type of construction used and shall be to the National Grid SA approval.

4.1.8 Spread Footing Design, all overturning moments, vertical and horizontal loads shall be resisted by vertical bearing pressures and no passive lateral resistance of the soil shall be considered.

4.2 Loadings for Design

4.2.1 Considering the foundation as an extension of the substation equipment support structures or building structure, it is obvious that the foundation must resist the same loading that the structures resist. Analysis on the structural system and the appropriate combination of axial, shear and moment loads acting on every foundation for each loading case based on the information specified in TES-P-119.09 and TES-P-119.19 shall be done by the Design Engineer. All other applicable loadings shall also be considered.

4.2.2 For substation structures, the Design Engineer shall analyze the loading combinations acting on the foundations for each loading case based on the criteria specified in TES-P-122.05 and TES-P-119.19.

a. Wind Loading:

i. 10 m high and below ---------- 1.00 kN/m^2
ii. 10 m up to 30 m---------- 1.25 kN/m^2
iii. Above 30 m---------- 1.50 kN/m^2
iv. Round surfaces---------- 0.60 kN/m^2

b. Imposed Loads:

All floors and roofs shall be designed to carry the Uniformly Distributed Load (U.D.L) or the Concentrated Load on an area whichever produces the greater stresses.

In addition to all dead loads as mentioned in TES-P-119.19, the structures shall be designed for any live loads that might reasonably be imposed not less than the following:

i.	Shop & working areas----------	10.00 kN/m^2
ii.	Walkways & mezzanines--------	4.00 kN/m^2
iii.	Control rooms----------	7.50 kN/m^2
iv.	Office & personnel spaces-------	3.50 kN/m^2
v.	Roofs	
	Without access----------	1.00 kN/m^2
	With access----------	1.50 kN/m^2

c. Equipment Loads:

Equipment loads shall be determined by the weight of the manufacturer's equipment supplied and by the dynamic or impact effects in the case of switchgear and rotating machinery.

d. Temperature Effects or Thermal Effects:

Temperature or Thermal effect shall be considered on structural design caused by the climate based on the Table–I below.

TABLE – I Mean Temperature and Temperature Variation °C

Material	Exposed to Sun	In Shade
Reinforced Concrete	30	25
Steel	30	25

In addition, where a structural element is exposed to a different extent on opposite faces (e.g., a column with one face outside and the opposite face inside a building) consideration shall be given to the thermal effects caused by the resulting temperature gradient across the structural section. In the absence of other data a temperature difference of 15 °C shall be considered in such cases.

The table of mean temperatures above may not necessarily give the maximum surface temperature conditions and reference should be made to the shade temperatures. Consideration shall also be given to the degree of exposure during construction.

Allowance shall be made for the stresses induced in any structure due to differences in temperature between the surfaces of a member or between parts of the structure.

The temperature of a concrete or steel surface exposed to sunshine may be taken as 70°C maximum.

e. Seismic loads

Foundations shall be designed as tabulated below in Table-II with corresponding seismic zone probability of Uniform Building Code.

TABLE-II SEISMIC ZONE TABULATION**

LOCATION	SEISMIC ZONE
HAFAR AL-BATIN	I
DHAHRAN	I
JEDDAH	IIA
KHAMIS MUSHAYT	I
RIYADH	0*

* Shall conform to the requirement specified in TES-P-119.19, latest revision.

** For other areas, refer 01-TMSS-01, Latest revision.

f. Crane Loads

All crane ways shall have their design load increased for the impact as follows:

1. A vertical force equal to 25% of the maximum wheel load.
2. The lateral force 20% of the weight of the trolley and lifted load only, applied one-half at the top of each rail.
3. The longitudinal force of 10% of the maximum wheel loads of the crane applied at the top of the rail.

g. Construction Loads

Consideration shall be given to the construction sequence in order to ensure that a structure is able to carry the design loads which act before completion of the structure and, also, that exceptional loads resulting from the construction sequence are catered for. This consideration shall include dead, wind and thermal loads together with suitable imposed loads, as appropriate.

4.2.3 Combination of Loads

All parts of a structure shall be designed for the worst combination of dead, imposed and equipment loads with temperature, wind, seismic and other effects. The following load combinations shall be considered:

D.L. + I.L.

D.L. + W.L.

D.L. + I.L. + W.L.

D.L. + S.L.

D.L. + I.L. + S.L.

where: D.L. = dead load

I.L. = imposed load

W.L. = wind load

S.L. = seismic load

The imposed load shall include for dynamic effects where appropriate. The imposed and wind loads shall be combined with the dead load and with each other in such a way as to give the worst possible stresses at any location. In addition thermal effects shall be considered, where significant, acting in conjunction with the combined load cases. Load cases need not be combined arbitrarily without regard for practical possibilities.
Foundation shall be designed using the reactions imposed by the structures they support.

4.3 Safety Factors

4.3.1 The factor of safety applied to the foundation design shall be greater than the factor of safety applied to the structure design. The overall factor of safety shall not be less than 3.0 against the maximum anticipated pressure applied to the soil.

4.3.2 Foundation design shall be based on the allowable bearing capacity of the soil if working loads (un factored) are to be considered.

4.3.3 The factor of safety against sliding shall not be less than 1.5. For overturning during construction or erection and other non-recurring load elements and for all other conditions, the factor of safety shall not be less than 2.0 based on working loads. Passive earth pressure from backfill shall be considered in computing these factors of safety. Unless specified otherwise in the detailed soil investigation, the coefficient of friction use in computing the safety factor against for cast-in-place foundations shall be 0.40.

4.3.4 For shallow foundations against uplift, factor of safety shall not be less than 2.5.

4.3.5 Unless otherwise established by the soil analysis or actual pile load test, the minimum safety factor, against the long-term operating loads, shall not be less than 3.0 for both compression and uplift reactions.

4.3.6 The factor of safety for bearing pressure for dead loads against soil bearing capacity shall be at least 3.0, and for combined dead and live loads, the factor of safety shall be at least 2.0.

5.0 SOIL INFORMATION

5.1 Soil Conditions in Saudi Arabia

The significant soil conditions prevalent in Saudi Arabia include the following:

5.1.1 Sabkha area wherein the material composition is firm enough to support roadways when dry, but becomes soft and unstable when water table rises. In case the substation location is on this type of soil, the development of Sabkha soil described in TCS-Q-113.02 shall be followed.

5.1.2 Areas of Eolian sand create the problem of shifting of the overburden at the foundation due to wind action. This shall be solved by stabilizing the surface with any method approved by National Grid SA

5.1.3 Areas of marl, wherein the material properties are cohesive in nature, are generally suitable for substation foundations. However, National Grid SA approval is required prior to its use.

5.1.4 Areas of rock outcroppings, where the rock is generally a weathered limestone that is drillable to a sufficient depth to install drilled piers or rock-anchor type of foundations.

5.2 Soil Classification

To promote uniformity in describing and interpreting descriptions of soil and their properties, and for reviewing soil data accumulated in a test boring program, refer to ASTM D2487.

5.3 Soil Testing

5.3.1 Of utmost importance in the design of foundations is an accurate and thorough knowledge of the site soil characteristics. Soil boring and testing programs are required and shall be undertaken at an early date.

5.3.2 The relevant standards shall be utilized to promote uniformity in obtaining test data on soil samples obtained in test boring programs.

5.3.3 For all foundations, site soil parameters must be ascertained from soil borings and from the laboratory soil tests. Soil properties and parameters listed below are commonly used in various design procedures:

a. Unit weight of the soil, kN/m^3 (γ).

b. Cohesive strength of the soil, in kN/m^2 (The cohesive strength is taken as one half of the unconfined compressive strength.)

c. The angle of internal friction of the soil, in degrees (ϕ)

d. The maximum expected elevation of the water table, in mm, (projected from water encountered during borings), the season of the year, and the topography of the area

e. The atterberg limits of cohesive soils

f. Grain size analysis of granular soils

g. Chloride and Sulfate tests for water and soil

h. Consolidation test

i. Electrical and Thermal resistivity tests

j. California Bearing Ratio (Soaked-CBR)

k. Unconfined strength of intact rock core, in case of rock

l. Submerge and saturated weight of soil

m. Modulus of subgrade reaction

n. Coefficient of lateral earth pressure.

o. Rock Quality Designation (RQD)

p. Cone Angle for uplift.

q. Standard Penetration Test (SPT) “N” values.

r. Dutch Cone Penetration Test “Rp” values.

5.3.4 Before carrying out soil investigation, the proposal for the soil geotechnical investigation shall be submitted to National Grid SA for review and approval. Soil investigation report shall be submitted to National Grid SA for review and approval.

5.3.5 Based on the specified foundation types, the recommendation for foundation selection shall be made by Geotechnical Investigation Agency acceptable to National Grid SA.

5.4 Soil Bearing Values

For Spread/Isolated footing and mat foundation, allowable bearing pressure shall be limited by the following considerations;

a. The factor of safety against ultimate shear failure must be adequate.

b. Settlement under allowable bearing pressure shall not exceed the tolerable values.

Ultimate bearing capacity shall be analyzed and computed. Allowable (Safe) bearing pressure shall be calculated with safety factor of not less than 3.0.

For deep foundations such as drilled piers, ultimate bearing capacity in cohesive and cohesionless soils shall be analyzed and computed.

5.5 Ground Water Table

Ground water table level is essential in foundation design because excavation below it requires dewatering which increases the cost. It has a significant effect on the bearing capacity of shallow foundations. The submergence of soil causes loss of all apparent cohesion and at the same time the unit weight of submerged soil will be reduced to one half the weight of the same soil above the water table. Thus, bearing capacity shall be analyzed using the highest possible groundwater level derived from the Geotechnical Investigation reports.

5.6 Differential Settlement

Clay, silt and silty sands are usually problematic types of soil. Weak strata of soil under a thin layer of dense or good soil shall be carefully examined and taken into consideration. To determine differential settlement between foundations that would cause damage to support equipment, settlement shall be investigated and reports submitted to National Grid Saudi Arabia approval. Foundations shall be sized so that settlement under dead load will not exceed 6.0 mm. Differential settlements of foundations for interconnected machinery is not acceptable. Foundations for buildings shall be designed so that differential settlements will not produce excessive stresses in the structural members or cause cracks in the walls.

5.7 Ground Improvement

Any measures taken to improve ground conditions shall be as per latest revision of TCS-Q-113.02, and in-situ tests shall be conducted to verify the improvement and results submitted for review.

6.0 TYPE OF FOUNDATIONS

The types of foundations mostly used for substation building structures and equipment support structures can be summarized as follows:

6.1 Drilled Piers

Drilled Piers used to support light equipment and structures having primarily horizontal and uplift loads such as bus support and gantry structures.

6.2 Spread/Isolated Footings and Mat Foundations

Spread/Isolated Footings and Mat Foundations are usually preferred for the foundation of control building and large electrical equipment.

6.3 Pad Foundations

Pad Foundations are sometimes recommended as foundation for miscellaneous equipment, and power transformer.

6.4 Driven Piles

Driven Piles maybe required where adequate bearing or uplift strength cannot otherwise be obtained.

7.0 DESIGN PROCEDURES

Given below are guidelines for the design of different types of foundations. The designer shall adopt clear principles and good judgment for the best designs as per International Standards mentioned in Section 3.0.

7.1 Drilled Piers

7.1.1 Top of concrete shall be 150 mm to 450 mm above final grade unless otherwise noted. Design reveal shall be 450 mm maximum.

7.1.2 Top of concrete of the foundations shall be at a constant elevation throughout the substation.

7.1.3 Pier embedment lengths shall be given in 300 mm intervals.

7.1.4 Concrete design procedure shall be in accordance with the requirements of ACI 318.

7.1.5 Foundations or footings with lateral loads shall be analyzed separately and submitted to National Grid SA for review and verification with all the calculations.

7.1.6 Drilled Piers with downward and upward loads shall be analyzed in accordance with relevant International Standards.

7.1.7 Concrete cover shall be 85 ± 10 mm, measured from the concrete surface of the drilled pier to the outside surface of the circular ties. (This minimum cover allows for some hole out-of-roundness and/or out-of-plumbness which may occur during construction, and for some flexibility of the reinforcing cage).

7.1.8 In designing pier reinforcement, axial loads, both tension and compression, shall be considered to extend the full length of the pier. The effect of skin friction in diminishing the axial load shall be ignored.

7.1.9 Shear reinforcement requirements, if any, will be determined from the maximum shear in the foundation. If shear reinforcement is required, it shall be provided for the full length of the pier.

7.1.10 Reinforcement Details

a. Minimum longitudinal reinforcement shall be 0.005 times the full cross sectional area of the pier. Additional reinforcement maybe required by bending in the pier. Not less than 16 mm bar size shall be used in longitudinal reinforcement and not less than four (4) bars shall be used in any pier.

b. Minimum ties shall be 10 mm diameter bars. Additional ties may be required by shear in the pier. Maximum spacing of ties shall not exceed 16 longitudinal bar diameters, or 48 tie diameters or 610 mm whichever is less.

c. Lap splice lengths have been calculated as 1.7 x the development length for longitudinal steel and 1.7 x development length x 0.8 for tie steel. Lap length for reinforcing steel shall be as indicated in Table-III.

TABLE – III Steel Lap Length

Rebar	Rebar Yield Strength For 420 MPa	
Size/Diameter (mm)	Tie Lap (mm)	Vertical Steel (mm)
10	432	-
12	432	-
16	533	660
20	686	840
25	1220	1500
28	-	1900
32	-	2410
36	-	2970

Note: Values listed in Table III are based on Concrete Strength of 21 MPa.

d. Vertical bars shall be placed within the circle of the ties, and wired to the ties at each intersection. Two (2) Tie wire shall be a minimum of 1.50 mm diameter.

e. Longitudinal steel shall extend to 85 ± 10 mm below top of concrete, and 85 ± 10 mm above the design bottom of pier. If during construction the pier is slightly longer than detailed, the rebar cage shall be lifted to within 75 mm of the top of the pier.

7.2 Spread/Isolated Footings and Mat Foundations

Design procedure and analysis shall be as per ACI 318.

7.2.1 The bearing capacity shall be determined as recommended in the soil investigation report.

7.2.2 Uplift capacity for spread footings shall be analyzed for the design.

7.2.3 The design of mat foundations shall be based on the theory for beams or plates on elastic support.

7.3 Pad Foundations

Pad foundation maybe subjected to jacking loads, dragging and impact loads, such as transformer pad. These loads shall be checked with all possible eccentricities.

7.4 Driven Piles

Driven piles are recommended for foundations located in areas with poor soil properties.

Piled foundations shall be designed in accordance with the solid information obtained from the Geotechnical Site Investigation.

The maximum compressive stresses in concrete piles shall not exceed 7 MPa. The calculated pile capacities shall be verified by load tests on selected piles. Ultimate pile bearing capacities and final sets for driven piles shall be based on the Hiley formula or a method using wave theory. For driven piles in cohesionless soils, estimated ultimate unit skin friction shall be based on the Geotechnical Soil Investigation Report. The ultimate pile bearing capacity shall be at least two and a half times the required pile working load.

Tolerance for piles, i.e., on plan position alignment and cut-off level, shall be specified on construction drawings. The effects of inaccuracies in the pile positions, within the tolerances specified shall be considered in the design of piled foundations.

7.5 Special Foundations

Special foundations (caissons, cofferdams, diaphragm walls, etc.) shall be designed using the soils information from the Site Soil Investigation. Any proposal for a new design/system shall be submitted to National Grid SA for review and approval.

8.0 MATERIALS

8.1 Portland Cement Concrete

Portland Cement Concrete shall be as specified in the latest revision of 70-TMSS-03.

8.2 Reinforcement

Reinforcing steel shall be deformed bars conforming to ASTM A615M, Grade 420 as required by the design. Reinforcement shall be free of loose or flaky rust and mill scale or coating, and any other substance that would reduce or destroy bond. Deformed hot-rolled high tensile steel bars having yield stress of 420 MPa shall in general be used as main reinforcements. Standard hooks shall be provided for mild steel bars.

9.0 REFERENCES

1. “The Ultimate Resistance of Rigid Piles Against Transversal Forces", Author J. Brinch Hansen

2. "Foundation Design and Construction", Author M. J. Tomlinson

3. NAVFAC DM-7, "Design Manual-Soil Mechanics, Foundations, and Earth Structures", Naval Facilities Engineering Command (Department of Navy)

4. REA Bulletin 65-1, "Design Guide for Rural Substations", US Department of Agriculture

5. "Foundation Design", Author W. C. Teng

6. “Foundation Analysis and Design”, Author Joseph E. Bowles